

Environmental Screening in MAAT Structural Cosmology

Emergent Suppression of Metric Response in High-Coherence Regions

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Abstract

Paper 45 introduced a minimal metric-response layer for MAAT structural cosmology by coupling the bounded projection kernel $\hat{\mathcal{C}}_{\text{proj}}(z)$ to both the effective growth response $\mu(z)$ and the gravitational-slip channel $\eta_{\text{slip}}(z)$. This paper extends that construction by making the projection response environment-dependent.

We define an environmentally screened projection kernel

$$\hat{\mathcal{C}}_{\text{env}}(z, \Delta, \Sigma_{\text{env}}) = \hat{\mathcal{C}}_{\text{proj}}(z) \mathcal{S}_{\text{env}}(\Delta, \Sigma_{\text{env}}),$$

where $\Delta = \rho/\bar{\rho}$ is a dimensionless density contrast, Σ_{env} is a dimensionless structural-stability or shear proxy, and

$$\mathcal{S}_{\text{env}} = [1 + \alpha_{\rho}\Delta_+^n + \alpha_{\sigma}\Sigma_{\text{env}}^m]^{-1}, \quad \Delta_+ = \max(\Delta - 1, 0).$$

The effective growth and metric-response channels become

$$\mu_{\text{env}} = 1 + \eta_g \hat{\mathcal{C}}_{\text{env}}, \quad \eta_{\text{slip,env}} = 1 + \beta_s \hat{\mathcal{C}}_{\text{env}}, \quad \Sigma_{\text{lens,env}} = \mu_{\text{env}} \frac{1 + \eta_{\text{slip,env}}}{2}.$$

Using the Paper-45 representative response parameters $(\eta_g, \beta_s) = (0.02, 0.03)$, we evaluate six synthetic environments: void, sheet, field, filament, cluster, and local-dense. The response is nearly unscreened in void-like regions ($\mathcal{S}_{\text{env}} \simeq 0.998$, $\max |\Sigma_{\text{lens}} - 1| \simeq 3.515\%$), suppressed in cluster-like environments ($\mathcal{S}_{\text{env}} \simeq 0.156$, $\max |\Sigma_{\text{lens}} - 1| \simeq 0.544\%$), and practically removed in a high-density local-test proxy ($\mathcal{S}_{\text{env}} \simeq 2.1 \times 10^{-4}$, $\max |\Sigma_{\text{lens}} - 1| \simeq 7.4 \times 10^{-4}\%$).

The result is an effective screening benchmark, not a microscopic chameleon, Vainshtein, or symmetron derivation. Its contribution is to show how a cosmological MAAT metric response can be structurally suppressed in high-density or high-coherence environments while remaining potentially visible in void-like regions.

1 Purpose and Status

Modified-gravity and dark-energy frameworks face a recurring consistency problem: cosmological deviations from general relativity may be interesting, but local and high-density environments are tightly constrained. Successful theories therefore often require some form of screening, such as chameleon, Vainshtein, or symmetron-type suppression mechanisms.

The purpose of this paper is not to derive such a microscopic mechanism. Instead, we ask a more limited question:

Can the MAAT projection response introduced in Papers 43–45 be suppressed by environmental structure in a bounded and reproducible way?

The answer developed here is an effective environmental screening layer. It should be read as a toy benchmark and a bridge toward future environment-dependent perturbation studies, not as a local-gravity test or a complete modified-gravity model.

2 From Metric Response to Environmental Response

Paper 45 used the bounded projection kernel $\hat{\mathcal{C}}_{\text{proj}}(z)$ to define the growth and slip responses

$$\mu(z) = 1 + \eta_g \hat{\mathcal{C}}_{\text{proj}}(z), \quad (1)$$

$$\eta_{\text{slip}}(z) = 1 + \beta_s \hat{\mathcal{C}}_{\text{proj}}(z), \quad (2)$$

$$\Sigma_{\text{lens}}(z) = \mu(z) \frac{1 + \eta_{\text{slip}}(z)}{2}. \quad (3)$$

Here μ controls the matter-growth source term, $\eta_{\text{slip}} = \Phi/\Psi$ controls the relation between scalar metric potentials, and Σ_{lens} parameterizes the Weyl/lensing response channel.

Paper 46 replaces the global projection kernel by an environmental one:

$$\boxed{\hat{\mathcal{C}}_{\text{env}}(z, \Delta, \Sigma_{\text{env}}) = \hat{\mathcal{C}}_{\text{proj}}(z) \mathcal{S}_{\text{env}}(\Delta, \Sigma_{\text{env}}).} \quad (4)$$

The screening factor is

$$\boxed{\mathcal{S}_{\text{env}}(\Delta, \Sigma_{\text{env}}) = [1 + \alpha_\rho \Delta_+^n + \alpha_\sigma \Sigma_{\text{env}}^m]^{-1}, \quad \Delta_+ = \max(\Delta - 1, 0).} \quad (5)$$

The density variable $\Delta = \rho/\bar{\rho}$ is dimensionless. The variable Σ_{env} is a phenomenological environmental-stability proxy. In this first benchmark it should be read as a normalized measure of structure, shear, or local coherence that suppresses the response when the environment is strongly organized.

3 Screened Growth and Metric Response

The screened response channels are

$$\mu_{\text{env}}(z) = 1 + \eta_g \hat{\mathcal{C}}_{\text{env}}(z, \Delta, \Sigma_{\text{env}}), \quad (6)$$

$$\eta_{\text{slip,env}}(z) = 1 + \beta_s \hat{\mathcal{C}}_{\text{env}}(z, \Delta, \Sigma_{\text{env}}), \quad (7)$$

$$\Sigma_{\text{lens,env}}(z) = \mu_{\text{env}}(z) \frac{1 + \eta_{\text{slip,env}}(z)}{2}. \quad (8)$$

Because $0 \leq \hat{\mathcal{C}}_{\text{proj}} \leq 1$ and $0 \leq \mathcal{S}_{\text{env}} \leq 1$, the environmental response is bounded by construction.

The matter-growth equation remains the Paper-43/Paper-45 equation, but now with the environmental growth coupling:

$$\frac{d^2 D}{dN^2} + \left[2 + \frac{d \ln H}{dN} \right] \frac{dD}{dN} - \frac{3}{2} \Omega_m(a) \mu_{\text{env}}(a) D = 0. \quad (9)$$

For $\mathcal{S}_{\text{env}} \rightarrow 0$, the model returns to the Λ CDM growth and metric-response limit:

$$\boxed{\mathcal{S}_{\text{env}} \rightarrow 0 \implies \mu_{\text{env}} \rightarrow 1, \quad \eta_{\text{slip,env}} \rightarrow 1, \quad \Sigma_{\text{lens,env}} \rightarrow 1.} \quad (10)$$

For $\mathcal{S}_{\text{env}} \rightarrow 1$, the model recovers the unscreened Paper-45 response.

4 Environmental Archetypes

The benchmark uses six synthetic environments. They are not empirical measurements. They are chosen to span a controlled hierarchy from low-density voids to high-density local-test proxies.

Table 1: Synthetic environmental archetypes.

Environment	Δ	Σ_{env}	Interpretation
void	0.20	0.05	underdense, weakly screened
sheet	0.80	0.15	mildly underdense sheet
field	1.00	0.20	mean-density field
filament	5.00	0.45	moderately overdense structure
cluster	100.00	0.85	dense cluster-like environment
local-dense	10^6	0.95	high-density local-test proxy

The screening parameters used throughout are

$$\alpha_\rho = 0.15, \quad \alpha_\sigma = 1.0, \quad n = 0.75, \quad m = 2.0. \quad (11)$$

The response parameters are inherited from the representative Paper-45 branch:

$$\eta_g = 0.02, \quad \beta_s = 0.03. \quad (12)$$

5 Results

All six environmental branches remain positive and stable in the minimal tested sense:

$$\mu_{\text{env}} > 0, \quad \eta_{\text{slip,env}} > 0, \quad \Sigma_{\text{lens,env}} > 0, \quad D > 0, \quad f > 0. \quad (13)$$

Table 2: Environmental screening results.

Environment	$\mathcal{S}_{\text{env}}$	$\max \mu - 1 $	$\max \Sigma_{\text{lens}} - 1 $	$\max$ Weyl-proxy deviation
void	0.9975	1.9916%	3.5151%	2.7245%
sheet	0.9780	1.9527%	3.4458%	2.6710%
field	0.9615	1.9198%	3.3873%	2.6259%
filament	0.6147	1.2274%	2.1592%	1.6766%
cluster	0.1555	0.3105%	0.5441%	0.4234%
local-dense	2.107×10^{-4}	0.00042%	0.00074%	0.00057%

The result is a monotonic environmental hierarchy: voids are almost unscreened, filaments are partially screened, clusters are strongly screened, and the local-dense proxy is effectively returned to GR.

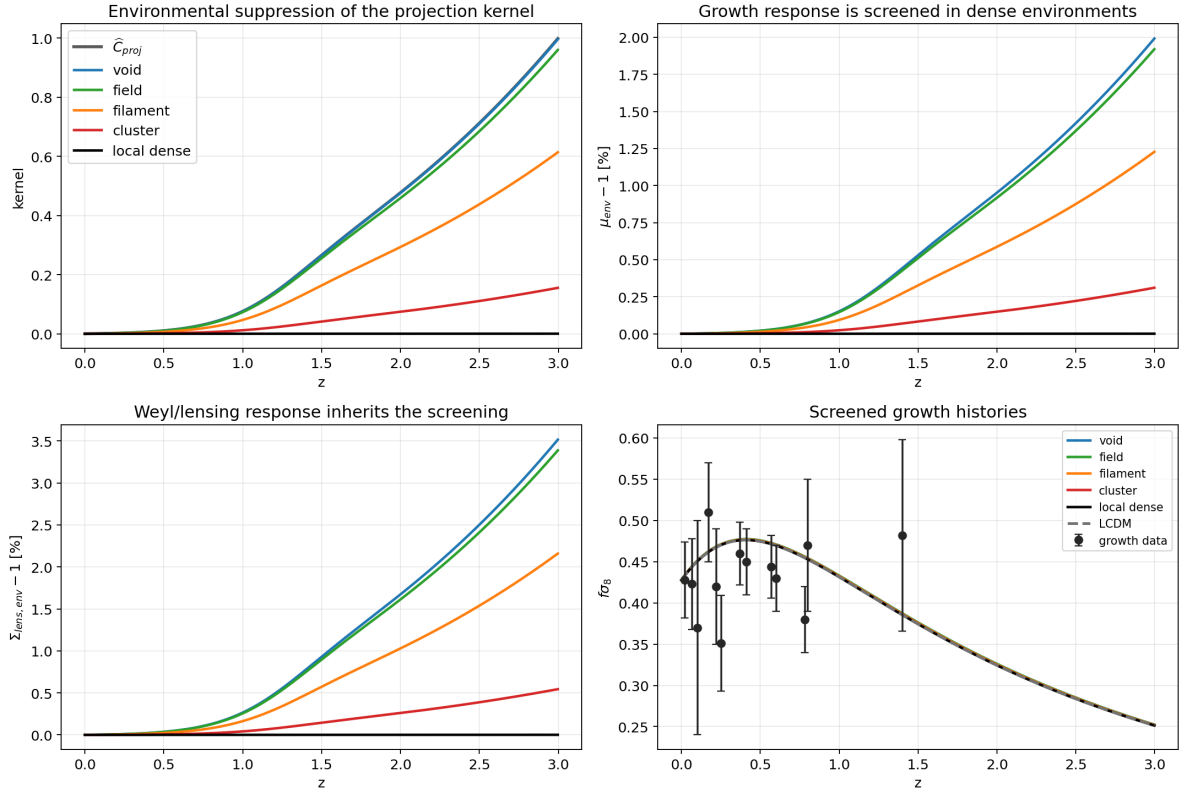


Figure 1: Paper 46 environmental screening summary. The same projection kernel produces different effective responses depending on environmental suppression.

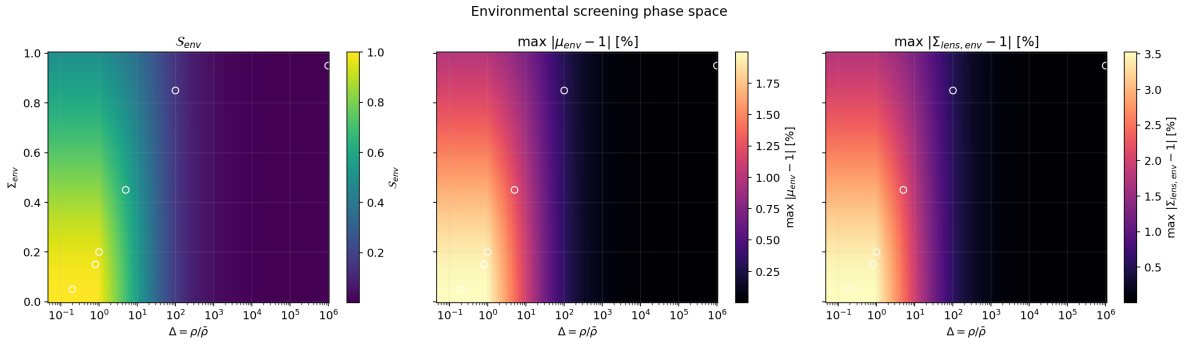


Figure 2: Screening phase space over density contrast Δ and environmental stability proxy Σ_{env} . High-density and high-stability regions suppress the response.

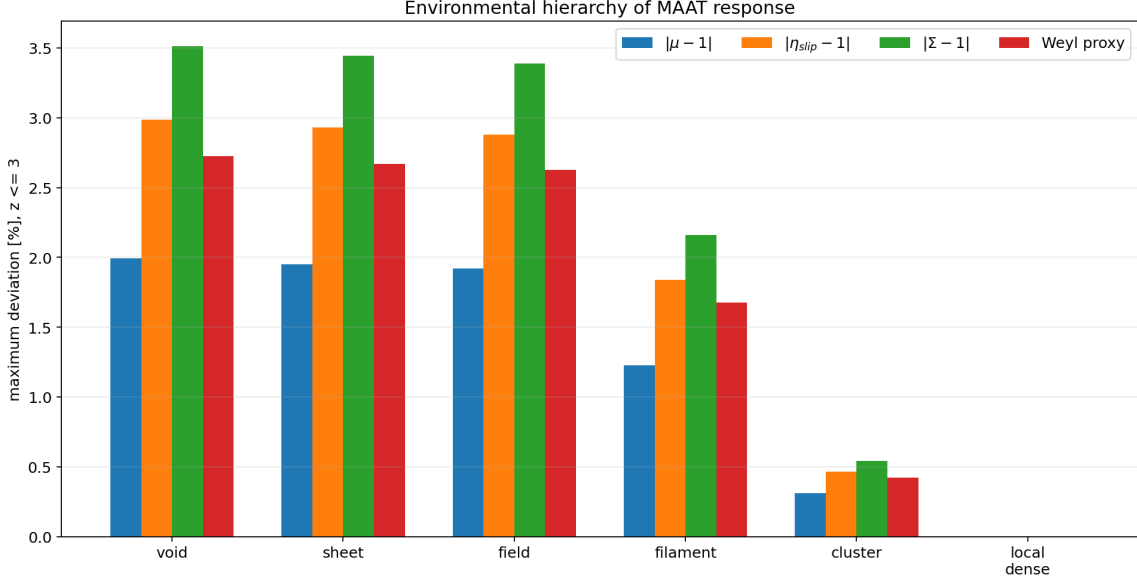


Figure 3: Environmental hierarchy of growth and metric-response deviations.

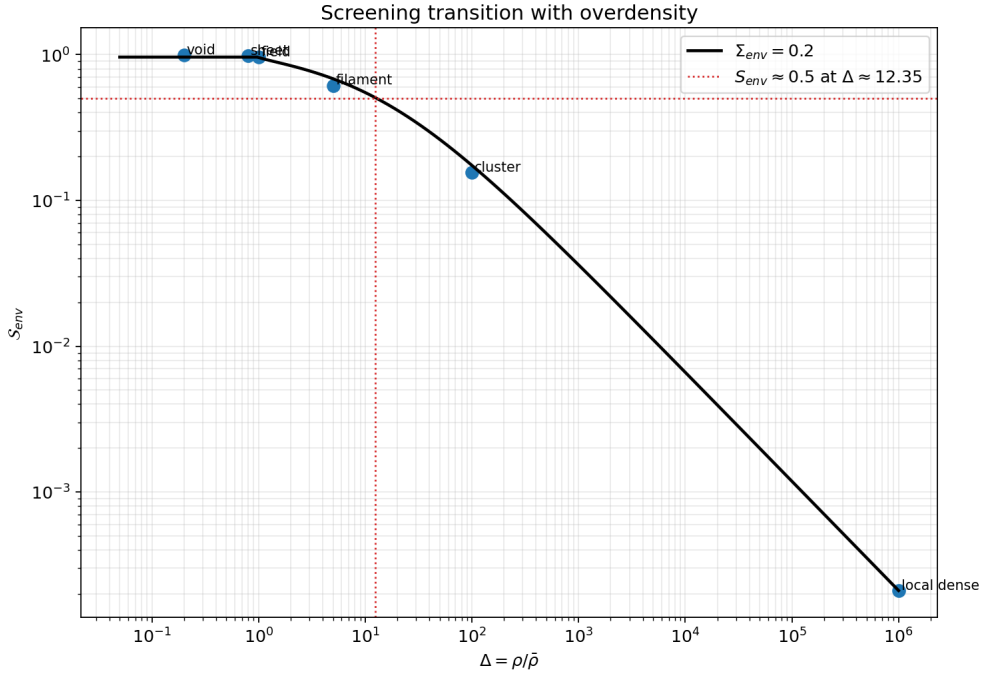


Figure 4: Screening transition with overdensity. For $\Sigma_{env} = 0.2$, the benchmark gives $\Sigma_{env} \simeq 0.5$ at $\Delta \simeq 12.35$.

6 Interpretation

The central result is:

MAAT metric response can be cosmologically active in low-density environments while being structurally suppressed in dense or highly organized regions.

This provides an effective answer to a standard concern for modified-gravity models. If a response is visible on cosmological scales, why is it not immediately excluded by high-density or

local tests? In the present construction, the answer is not an additional screening field. Instead, screening appears as environmental suppression of the existing projection response.

The benchmark also makes a clear observational distinction:

- voids should be the most sensitive regions for MAAT metric response,
- clusters should show strongly reduced deviations,
- local high-density environments should be close to GR,
- lensing/growth consistency should be environment-dependent.

7 Relation to Known Screening Mechanisms

The present construction is not a chameleon, Vainshtein, or symmetron model. Those mechanisms introduce specific field dynamics, nonlinearities, or environment-dependent effective masses. Here, by contrast, the response is suppressed directly at the level of the structural projection kernel.

The relationship is therefore conceptual rather than microscopic:

standard screening suppresses a field-mediated force; MAAT screening suppresses the structural response kernel that feeds growth and metric response.

8 Limitations

Several limitations are important:

- Σ_{env} is phenomenological and not yet derived from a shear tensor, halo model, or density field.
- The environment variables are synthetic archetypes, not survey data.
- No scale dependence $\mu(k, z, \Delta)$ or $\Sigma(k, z, \Delta)$ is implemented.
- No N-body, halo-model, weak-lensing, or cluster-likelihood analysis is performed.
- The benchmark is not a local-gravity test and does not replace Solar System constraints.
- The screening law is an ansatz, not a first-principles derivation.

9 Reproducibility

The code and outputs are stored in:

`experiments/maat_environmental_screening_paper46/`

Run:

```
python3 maat_environmental_screening_solver_v01.py
```

The script writes:

- `outputs_paper46/paper46_summary.json`,
- `outputs_paper46/paper46_environment_archetypes.csv`,
- `outputs_paper46/paper46_environment_curves.csv`,

- `outputs_paper46/paper46_screening_grid.csv`,
- four PNG figures used in this paper.

The public repository is:

<https://github.com/Chris4081/structural-selection-principle>

Data attribution and license note. The Planck-normalised reference parameters and compact $f\sigma_8$ comparison points are external scientific data and should be cited to the original publications/collaborations when reused or discussed. The CSV tables and figures in this repository are derived reproducibility artifacts only. No endorsement by the Planck Collaboration, survey collaborations, or original data authors is implied.

10 Conclusion

We introduced an environmental screening layer for MAAT structural cosmology. The Paper-45 projection response is multiplied by a dimensionless environmental suppression factor depending on overdensity and structural stability. This yields an effective transition from near-unscreened void response to almost complete high-density suppression.

The result is not a microscopic screening theory. It is a reproducible effective benchmark showing how MAAT metric response can be made environment-dependent in a way that is qualitatively compatible with the need for cosmological visibility and local suppression.

References

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